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The Petrology and Paleomagnetism of the Coronation Sills[1]

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The Coronation Sills and their extrusive equivalents of the Franklin Magnetic Division form a major basaltic province of Hadrynian age in the northwestern part of the Canadian Shield. The sills are saturated or slightly undersaturated, undifferentiated tholeiites. They are similar in composition to tholeiitic plateau basaltic provinces elsewhere, particularly to the Karroo Province. Most of the paleomagnetic samples contain secondary components that are much reduced by magnetic cleaning.

The optimum cleaning field determined by the precision method is 105 Oe, but by the end-point method it is in the 200–400 Oe range. Sites may be separated by thermal cleaning into a stable group with a dominant discrete blocking temperature above 500°C, and a less stable group with blocking temperatures distributed in the 50–550 °C range. The pole position calculated from samples cleaned in 210 Oe is 1° S, 163° E ($\alpha_{95} = 9°$). The position of the revised pole for the Franklin Magnetic Interval, incorporating the Coronation Sills data, is at 5° N., 168° E. ($\alpha_{95} = 5°$).

Les filons couche Coronation et leurs equivalent extrusif de la division magnétique Franklin appartiennent à une province basaltique majeure d'âge hadrynien, au nord-ouest du bouclier canadien. Les filons couche sont constitués de tholeites saturées ou légèrement sous-saturées. Elles ont des compositions semblables aux basaltes tholeitiques des plateaux d'autres provinces, le Karroo par exemple. La plupart des échantillons paléomagnetiques contiennent des constituants secondaires en partie éliminés par désaimanation.

Le champ de désaimantation optimum déterminé par la méthode précision est 105 Oe, mais il est de 200 à 400 Oe si determiné par la méthode "end-point".

Les localités peuvent être séparées par désaimantation en un groupe stable ayant une température de blocage spécifique supérieure à 500° C et un groupe moins stable avec des températures de blocage allant de 50 à 500° C. La position du pôle calculée à partir d'échantillons désaimantés dans 210 Oe est 1° S, 163° E ($\alpha_{95} = 9°$). La position révisée du pôle pour l'Intervale Magnétique Franklin, comprenant les données sur les filons couche Coronation, est à 5° N, 168° E. ($\alpha_{95} = 5°$).

Introduction

The Coronation Sills, together with the Natkusiak basalts, form a major basaltic province in the northwestern part of the Canadian Shield. They derive their name from Coronation Gulf (Baragar and Donaldson, in press), where they are prominantly displayed along the coast and on islands. This study concerns the sills both petrographically and paleomagnetically, and new data are presented from both aspects. It is a detailed study of a major intrusive unit in the Franklin Magnetic Division of Fahrig *et al.*

[1]Earth Physics Branch Contribution No. 365.

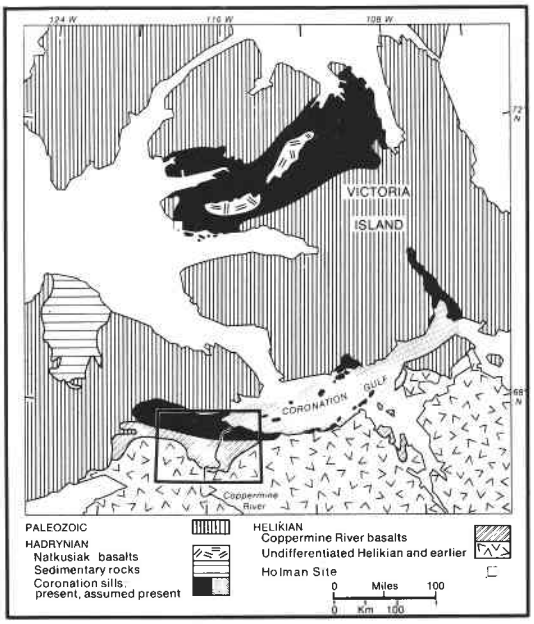


FIG. 1. Generalized geological map of the area showing the extent of the sills, the location of Fig. 2, and the paleomagnetic site on Victoria Island.

(1971). The sills were sampled west of Coppermine River where they form prominent cuesta ridges, ideal for collecting oriented cores by rock drill.

The objects of this study are therefore three fold: firstly to obtain chemical, petrographic and paleomagnetic data from the Coronation Sills; secondly to derive a pole position for them from stability tested data, and to combine the results with those of Fahrig *et al.* (1971) to update the Franklin Magnetic Interval Pole; and thirdly to use the Coronation Sills to compare the various factors that make up the stability spectra of these tholeiitic sheets. These factors may be usefully extrapolated to similar sheets that occur at many critical horizons in the geological column.

Geology and Age

The geological setting (Fraser 1960; Thorsteinsson and Tozer 1962; Christie 1964) and distribution of the Coronation Sills are shown in Fig. 1. The sills intrude gently dipping Hadrynian sediments on the northern fringe of continental North America, and Victoria Island, and underlie the intervening waterways. Paleozoic sediments cover the Hadrynian rocks over much of the region, but judging from the outcrop distribution the Coronation Sills underlie a circular area about 400 km in diameter.

On Victoria Island the sills have intruded sandstone, limestone, dolomite, and gypsum of the Shaler Group. Conformably overlying the Shaler Group (Fig. 1) are the plateau basalts of the Natkusiak Formation (Thorsteinsson and Tozer 1962), which Christie (1964) considered to be the extrusive equivalents of the sills. Christie (1964, p. 2) estimated the aggregate thickness of the sills in this region to be 450 to 800 m and that of the Natkusiak Volcanics to be about 300 m.

A more detailed map showing the mainland sampling sites is shown in Fig. 2. The geology is adapted from maps of the Coppermine and Dismal Lakes areas by Baragar and Donaldson (1970, and in press), and the new stratigraphic nomenclature is defined in the latter. Below is a generalized table of formations:

	Paleozoic dolomite
	Unconformity
Hadrynian	Coronation Sills
	Rae Group—dolomite, sandstone, siltstone, and shale
	Unconformity
Helikian	Coppermine River Group—red sandstones and intercalated basalt flows; basalt flows
	Dismal Lakes Group—dolomite, sandstone, mudstone, and shale
	Unconformity
	Muskox intrusive complex
	Undivided Helikian and Aphebian rocks

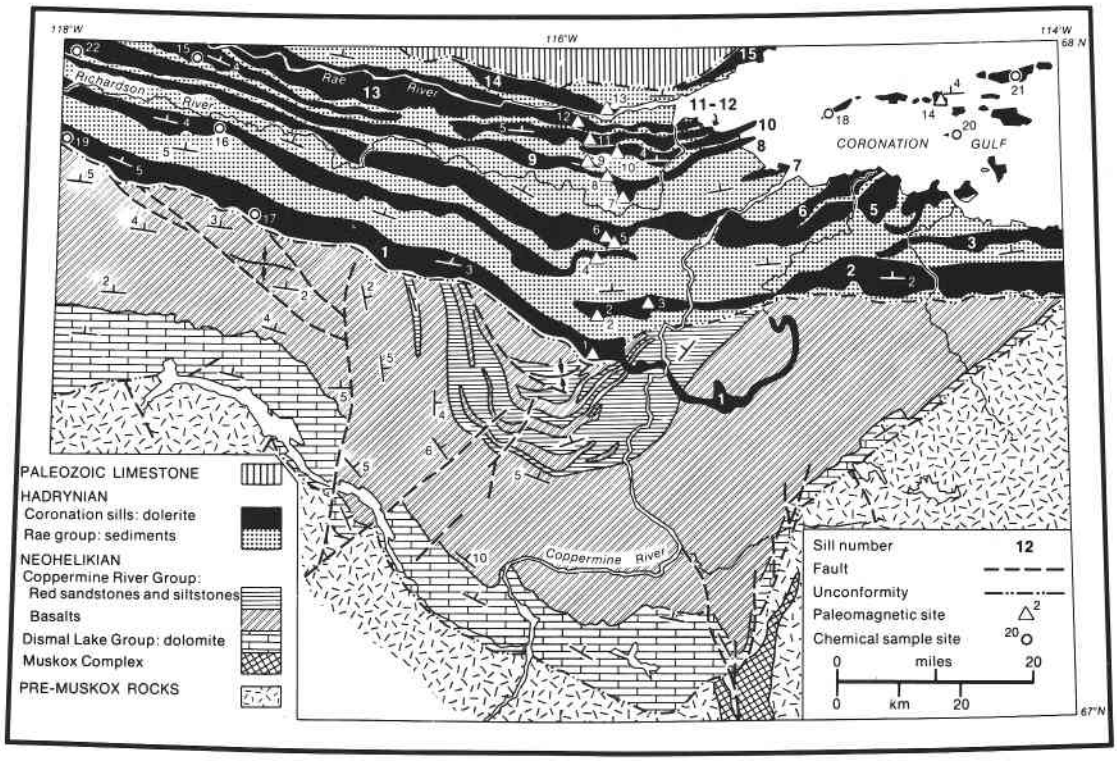


FIG. 2. Geological map of the area sampled in detail, showing the sills numbered up from the bottom, and the paleomagnetic (△) and chemical (○) sampling sites.

TABLE 1. K/Ar age determinations

Age (m.y.)	Locality	Reference
635	Victoria Island— 72°26′ N, 111°26′ W	Leech *et al.* 1963, p. 55; Christie 1964, p. 9.
640	Victoria Island— 72°26′ N, 111°26′ W	Leech *et al.* 1963, p. 55; Christie 1964, p. 9.
605 ± 110	Coronation Gulf area—68°03′ N, 118°20′ W	Wanless *et al.* 1966, p. 41.
718 ± 120	Coronation Gulf area—67°32′ N, 115°48′ W	Wanless *et al.* 1966, p. 48.
647 ± 32	Coronation Gulf area—67°33′ N, 116°23′ W	Wanless *et al.* 1970, p. 44.
631 ± 40	Coronation Gulf area—67°40′ N, 111°03′ W	Wanless *et al.* 1970, p. 46.
604 ± 92	Coronation Gulf area—67°47′ N, 117°41′ W	Wanless and Loveridge (in preparation).

All groups above the Muskox Intrusion dip northwards at angles up to 10°. The unconformity between the Coppermine River and Rae Groups is most evident where the broad warp in the Coppermine River Group is truncated northwards by the undeformed Rae Group sediments. Faults that offset the Coppermine River Group do not penetrate the Rae Group.

Fifteen separate sills are present within the mainland area sampled, but not all are continuous throughout. Most of the sills appear to be 50 to 100 m thick. Their aggregate thickness is estimated to be 1300 to 1500 m. However the very low dips make individual sill thicknesses difficult to measure and this estimate may be too high.

Potassium–argon age determinations of Coronation Sills are given in Table 1. The average age of 647 m.y. agrees well with the 650 m.y. average computed by Fahrig *et al.* (1971) from the most reliable determinations obtained from rocks intruded in the Franklin Magnetic Interval. Because of probable argon loss they prefer a mean age of 675 m.y., probably the best estimate for the time of intrusion of the Coronation Sills.

TABLE 2. Coronation Sills

(*a*) *Chemical analyses*

Sample No.	Site No.	Sill No.	SiO_2	Al_2O_3	Fe_2O_3	FeO	FeO_T	MgO	CaO	Na_2O	K_2O	TiO_2	MnO	P_2O_5	CO_2	S	H_2O	Fe_2O_3/FeO	Total
BL9–792	13	14	48.7	15.3	2.0	8.8	10.6	6.9	11.7	1.9	1.1	1.02	0.15	0.07	0.05	0.03	1.6	.23	99.3
BL9–788	12	13	50.9	13.2	3.9	12.7	16.2	4.2	8.1	2.8	0.9	2.73	0.27	0.22	0.10	0.08	1.9	.31	102.0
BL9–632	15	12A*	49.5	14.4	2.4	9.7	11.9	8.1	11.1	2.1	0.5	1.34	0.20	0.09	0.05	0.02	1.2	.25	100.6
BL9–670	21	12*	50.3	14.0	1.9	11.6	13.3	6.7	10.4	2.4	0.8	1.81	0.20	0.14	0.10	0.08	1.7	.16	102.1
BL9–784	11	11*	50.9	15.4	1.7	9.9	11.4	7.1	11.8	1.9	0.4	1.12	0.19	0.08	0.10	0.05	1.4	.17	102.0
BL9–780	10	10	48.9	15.7	2.9	8.3	10.9	6.2	11.7	2.3	0.2	1.25	0.19	0.10	0.05	0.01	1.1	.29	98.9
BL9–776	9	9*	49.4	15.3	2.0	9.2	11.0	7.9	11.6	2.2	0.4	1.26	0.17	0.09	0.05	0.01	1.1	.22	100.6
BL9–768	7	8*	49.2	12.9	1.7	11.9	13.4	6.4	10.6	2.5	0.6	1.81	0.21	0.13	0.05	0.03	1.1	.14	99.1
BL9–651	18	8*	50.2	13.9	1.4	11.1	12.4	8.4	9.5	2.4	1.0	1.78	0.19	0.14	0.05	0.03	1.3	.13	101.3
BL9–772	8	8	49.9	15.0	1.8	9.6	11.2	6.3	9.7	3.3	1.1	1.64	0.17	0.12	0.20	0.01	2.5	.19	101.3
BL9–683	22	8*	49.4	13.9	0.9	12.0	12.8	6.6	10.7	2.2	1.0	1.63	0.20	0.12	0.05	0.19	1.7	.13	100.5
BL9–760A	6	6	50.2	14.6	3.1	9.9	12.7	7.3	10.3	2.3	0.5	1.64	0.20	0.13	0.05	0.05	1.2	.31	101.4
BL9–667	20	6*	50.1	14.3	2.3	11.4	13.5	5.7	10.9	2.8	0.3	1.76	0.21	0.13	0.05	0.03	1.1	.20	101.0
BL9–764	5	5	50.1	15.9	2.7	9.8	12.2	5.4	10.1	2.7	0.7	1.90	0.19	0.17	0.05	0.02	1.4	.28	101.0
BL9–634	16	3	50.2	14.4	3.3	8.9	11.9	6.7	11.1	2.1	0.8	1.48	0.20	0.10	0.30	0.03	1.5	.37	101.1
BL9–750	2	2	50.6	14.6	2.3	10.6	12.7	7.0	10.6	2.3	0.6	1.57	0.20	0.11	0.05	0.01	1.0	.22	101.5
BL9–754	3	2	48.8	14.1	1.2	9.8	10.9	8.5	10.6	2.3	0.4	1.45	0.17	0.11	0.10	0.05	1.3	.12	98.9
BL9–643	17	1*	50.4	14.2	2.8	11.2	13.7	6.2	10.4	2.2	0.7	1.68	0.22	0.12	0.10	0.02	1.6	.25	101.8
BL9–679	19	1*	48.8	13.4	1.6	11.1	12.5	7.8	10.6	2.4	0.3	1.63	0.21	0.13	0.40	0.04	0.2	.14	98.6
Average			49.8	14.4	2.2	10.4	12.4	6.8	10.6	2.4	0.7	1.61	0.20	0.12	0.10	0.04	1.4	.22	

*Chilled contact of sill.

Analyses by Rapid Methods Groups, Analytical Laboratories, Geol. Surv. Can.

SiO_2, Al_2O_3, Fe_2O_3 Total, MgO, CaO, K_2O, TiO_2, MnO by X-ray fluorescence analyses; FeO, Na_2O, P_2O_5, CO_2, H_2O, and S by rapid chemical methods.

(*b*) *Cation norms*

Sample No.	Site No.	Sill No.	Q	Or	Ab	An	Di	He	En	Fs	Fo	Fa	Mt	Il	Ap	py	cc
BL9–792	13	14		6.7	17.6	30.9	14.3	8.3	8.6	4.9	3.0	1.7	2.2	1.5	.15	.08	.13
BL9–788	12	13	4.7	5.5	26.0	21.5	6.6	7.7	8.7	10.1			4.2	3.9	.48	.15	.26
BL9-632	15	12A		3.0	19.1	28.8	13.7	7.2	13.7	7.2	1.6	.9	2.5	1.9	.19	.05	.12
BL9–670	21	12		4.8	21.8	25.4	11.4	8.8	11.6	9.0	1.0	.8	2.0	2.6	.30	.21	.26
BL9–784	11	11	1.4	2.4	17.2	32.5	12.2	8.0	13.6	8.9			1.8	1.6	.17	.13	.25
BL9–780	10	10	1.2	1.2	21.4	33.1	13.5	7.4	11.0	6.0			3.1	1.8	.21	.03	.13
BL9–776	9	9		2.4	20.0	31.0	13.8	7.2	10.9	5.7	3.1	1.6	2.1	1.8	.19	.03	.13
BL9–768	7	8		3.7	23.3	23.1	13.1	11.2	9.3	7.9	1.8	1.6	1.8	2.6	.28	.08	.13
BL9–651	18	8		5.6	21.7	24.4	10.9	6.6	11.4	6.9	4.9	2.9	1.5	2.5	.30	.08	.12
BL9–772	8	8		6.6	30.1	23.2	11.3	7.5	2.9	1.9	6.8	4.6	1.9	2.3	.26	.03	.51
BL9–683	22	8		6.1	20.4	24.5	12.5	10.8	8.7	7.5	2.9	2.5	1.0	2.3	.26	.51	.13
BL9–760A	6	6	1.1	3.0	20.9	28.4	11.4	6.3	14.7	8.1			3.3	2.3	.28	.13	.13
BL9–667	20	6		1.8	25.6	26.1	11.8	10.5	9.6	8.4	.4	.4	2.5	2.5	.28	.08	.13
BL9–764	5	5	.8	4.2	24.7	29.8	9.2	6.7	10.6	7.7			2.9	2.7	.36	.10	.13
BL9–634	16	3	2.3	4.8	19.2	28.0	13.4	7.0	12.2	6.4			3.5	2.1	.21	.08	.77
BL9–750	4	2	.4	3.6	20.8	28.0	11.5	7.7	13.7	9.2			2.4	2.2	.23	.03	.13
BL9–754	3	2		2.4	21.2	27.7	13.0	6.9	11.6	6.2	4.5	2.4	1.3	2.1	.24	.13	.26
BL9–643	17	1	1.7	4.2	20.1	27.3	10.7	8.4	12.1	9.5			3.0	2.4	.26	.05	.26
BL9–679	19	1		1.8	22.1	25.6	12.1	7.9	12.5	8.2	2.6	1.7	1.7	2.3	.28	.11	1.0

Chemistry and Petrography

The sills are composed of fresh tholeiitic dolerite with minor deuteric alteration. The fine-grained margins contain about 10% of microphenocrysts of plagioclase and olivine and in places augite. The interior is medium grained, homogeneous, and sub-ophitic. It is composed of 40 to 45% each of plagioclase and pyroxene, 2 to 5% olivine and derived pseudomorphs, and 5 to 7% opaque minerals. Minor constituents include biotite, amphibole, and rare interstitial quartz. Low-calcium pyroxene was not recognized in thin section. Possibly, therefore, the augite is subcalcic and neither pigeonite nor orthopyroxene was precipitated.

Ade-Hall (1964) found a correlation between reversed magnetization and the pseudomorphing of olivine by iddingsite and secondary iron oxides in an olivine basalt succession in Mull. The inference is that the iddingsite-bearing rocks are self-reversed. He also correlated euhedral, early-crystallized magnetites with normal magnetization. The olivine in the Coronation Sills is largely altered to lamellar minerals with high birefringence that range in color from golden yellow to pale green. They have higher refractive indices than the saponite group and lack the color of iddingsite. Their identification is therefore uncertain. Secondary oxide minerals typically rim the olivine pseudomorphs and line the transverse fractures that cross them. The opaque minerals are subhedral, roughly cubic and may have developed quite early in the crystallization sequence. Since the altered olivine rarely comprises more than 5% of the Coronation Sills, against 10 to 20% in the Mull basalts, it is unlikely to have a significant effect on the paleomagnetic properties, and the reversed directions at all sites are considered to be caused by a reversal of the magnetic field, and not self-reversals.

Chemical analyses, mostly of lower chilled contacts of the sills, together with their norms, are given in Table 2. These analyses of rock from the fine-grained margin should represent the original magma of the sills.

The analyses are arranged in stratigraphic sequence, the lowest at the bottom. No trends are detectable. Sills should be emplaced successively upwards in a stratigraphic sequence[2], but there is no evidence in this sequence of significant differentiation during intrusion. This is surprising for so thick an aggregate magma succession (1300–1500 m), and may indicate that the whole sequence was intruded in a comparatively short time.

Chemical analyses in Table 2, and frequency distribution curves in Fig. 3, which are compared with curves for the nearby Coppermine River flows, show the characteristics of the Coronation Sills. The notable features are as follows:

1. The dolerites are distinctly tholeiitic, since all analyses show significant quantities of normative orthopyroxene. They are, however, saturated or slightly undersaturated, as shown by the very minor quantities of normative quartz and normative olivine appearing in about an equal number of analyses.
2. Compositional variation between the sills is small, indicating very minor differentiation during the course of intrusion. This is apparent in the very much narrower frequency distribution curves of the Coronation Sills (Fig. 3), compared with those of the Coppermine River flows, in which some differentiation took place during the period of extrusion.
3. The much higher oxidation state of the Coppermine River flows compared with the Coronation Sills is a reflection of their extrusion into the atmosphere. In other respects the two groups are similar except for distinctly higher lime and lower potash and titania in the sills.
4. It is evident from Table 3 that the Coronation Sills are generally similar to plateau basaltic provinces elsewhere (Coppermine, Karroo, Tasmania, Yakima), but markedly richer in potash than the tholeiitic basalts of geosynclinal assemblages (Timmins–Noranda, Labrador Trough).

[2]A sill will tend to form at a level in the magma column where the pressure exceeds that in the adjoining rock, plus a factor for resistance to intrusion. Since the density of basaltic magma is greater than that of sedimentary rocks the magma column will not reach the surface before the critical pressure is reached at the base of the sedimentary pile. A sill intruded at that level will raise the average density of the sedimentary sequence below but not above this level. Thus the emplacement of later intrusions will be favored at successively shallower levels.

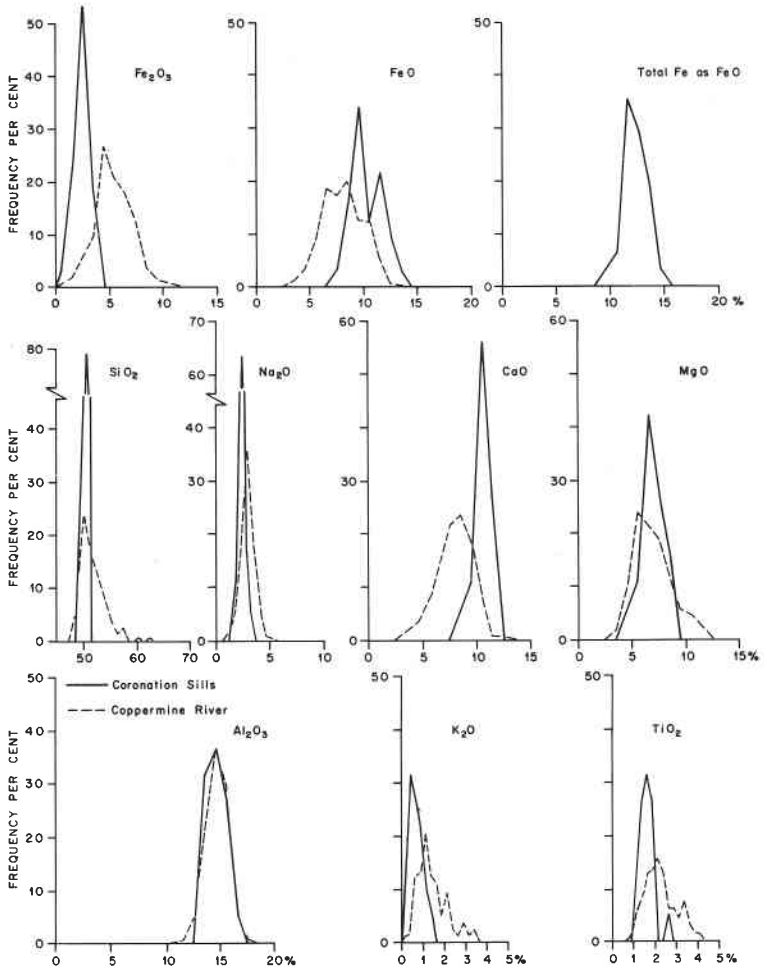


FIG. 3. Frequency curves showing the bulk chemistry of the Coronation Sills compared with that of the Coppermine River flows.

TABLE 3. Average chemical analyses of some basic igneous provinces[1]

	No. of analyses	SiO_2	Al_2O_3	Fe_2O_3	FeO	MgO	CaO	Na_2O	K_2O	TiO_2	P_2O_5	MnO
Coronation Sills	19	50.5	14.6	2.2	10.5	6.9	10.8	2.4	0.7	1.7	0.1	0.20
Coppermine River[2]	163	51.2	13.7	5.6	8.1	6.9	8.0	3.0	1.5	2.3	0.2	0.22
Karroo dolerites[3]	44	52.7	15.4	1.4	9.4	6.6	10.0	2.2	0.9	1.2	0.2	0.22
Tasmanian dolerites[4]	13	53.7	15.5	0.7	8.4	6.8	11.2	1.7	1.0	0.7	0.1	0.15
Yakima (Columbia)[5] basalts		54.5	14.3	2.6	9.3	4.2	8.0	3.0	1.5	2.0	0.4	0.20
Timmins–Noranda[6]	141	52.5	15.1	2.4	9.7	6.3	9.8	2.8	0.27	1.3	0.1	0.21
Labrador Trough[7]	22	50.6	14.6	2.5	10.2	7.5	10.4	2.6	0.23	1.1	0.1	0.23
Oceanic tholeiite[8]	25	50.0	17.1	2.1	6.9	7.6	11.7	2.8	0.18	1.4	0.1	0.16

[1]Analyses are all recalculated to 100 % water-free, CO_2-free.
[2]Baragar 1969.
[3]Walker and Poldervaart 1949.
[4]McDougall 1962.
[5]Waters 1961. Average of 4 composite samples representing a total of 55 flows.
[6]Baragar and Goodwin 1969. Recalculated to include basalts and type A basalts.
[7]Dimroth *et al.* 1970.
[8]Average of analyses reported by Nicholls *et al.* 1964; Engel and Engel 1963; Engel, Engel and Havens 1965; Engel, Fisher and Engel 1965; Cann and Vine 1966; Engel and Fisher 1969.

TABLE 4. Site rankings from stability criteria

I		II	III		IV	
Site	Arc (°)	Site	Site	k	Site	Arc (°)
10	9	4	9	179	10	2
9	15	8	8	107	11	8
1	25	3	7	101	9	20
6	26	11	3	100	1, 4	23
4	27	9	11	88		
11	30	5	10	61	2	36
8	32		13	60	8	38
		2, 13				
12	34		12	31	3, 13	46
3	39	6	4	19		
5, 13	40	1	2	9.4	7	47
		7	5		12	55
2	65	12	6	5.7	5	64
7	67	10	1	4.6	6	71

NOTE: I is the change in degrees in mean site direction from NRM to 210 Oe. II is the area under the demagnetization curves between 150 and 600 Oe. III is the maximum precision after cleaning 3 specimens; k is Fisher's (1953) best estimate. IV is the direction change in degrees between the mean site NRM direction and the mean formation direction after cleaning in 210 Oe.

They have a striking similarity to the Karroo dolerites.

Aspects of Magnetic Stability

Most rocks acquire, to varying degrees, secondary components at some time after they were formed. Commonly these have lower coercivity and blocking temperatures than the primary component and can be readily removed by cleaning techniques. The secondary components are then said to be unstable and the stable component reveals the true direction of the magnetic field at the time the rocks were formed.[3] Thermal and magnetic cleaning are the techniques most commonly used.

Thermal cleaning consists of heating specimens to a known temperature in a zero field (Irving *et al* 1961), whereas magnetic cleaning (Creer 1959) entails partial demagnetization of specimens in an alternating magnetic field of known peak value in a zero ambient field. The problem in each case is to devise satisfactory criteria for defining the best field or temperature to eliminate the secondary components. The methods usually used are either: (1) the end-point method (As and Zijderveld 1958; McElhinny and Gough 1963), which assumes the best field to be that at which the change in direction ceases; or (2) the precision method, in which the field is determined by the minimum dispersion of a group (Irving *et al.* 1961).

In Table 4 the 13 sites studied in detail using alternating field demagnetization have been ranked according to their stability using different criteria. In column 1 the criterion is the angular change in direction of the mean of 3 specimens between NRM and the same specimens cleaned in 210 Oe. Column 2 is ranked according to the area under the intensity demagnetization curves between 150 and 600 Oe. The order of column 3 is determined by the precision of the groups of 3 specimens at each site, Fisher's (1953) estimate of precision (k) being used to determine the order. In column 4 the difference in direction between the NRM site mean and the final direction of the formation is used to rank the sites.

The wide variation in ranking order by the different criteria appears to indicate that they are measuring different aspects of stability. However, some of the differences may be caused by inadequate data, for instance estimates of precision from a group of 3 are not very reliable; moreover, the demagnetization curve is from only one of the 3 specimens and may not be

[3]Weathered specimens are known to acquire a secondary component that cannot be removed without removing the primary component (Robertson and Hastie 1962); in such cases the primary component of magnetization is unobtainable.

TABLE 5. Opaque mineralogy

Site	% opaques	Size (in μ)	Shape	v. small opaques	Oxidation Index	Mag.-Il. intergrowths	Magnetite	Ilmenite	Hematite	Ulvöspinel	Spinel	Sulfides: Size range (in μ)	Sulfides: Type	Sulfides: Amount
P–1	5.4	100–1500	EA	**	C4	LX	•	•	•	*		10–20	Fs	•
P–2	2.4	100–1000	A–C	**	C3	LX	0	0	—		*	5–60	Fs	•
P–3	1.6	150–1000	A–C	**	C3	LX	•	0	—		*	5–60	Fs	•
P–4	4.1	150–1000	C–K	*	C3–C4	LX	•	0	•	*	*	5–60	Ch, Fs	•
P–5	5.3	100–2000	A	*	C2–C5	LX	•	(K) •	—	*	*	5–60	Fs	•
P–6	5.5	100–3000	EA–C	*	C3–C4	LX	•	(K) •	•	*	*	10–200	Fs	•
P–7	4.4	200–1200	C	**	C2–C3	LX	0	(K) •	—	*		5–60	Fs, Ch	•
P–8	3.3	100–2000	A–K	*	C3–C4	LX	•	•	•			10–120	Fs	•
P–9	2.9	200–1500	A–C	*	C4–C5	LX	0	•	•			5–100	Fs	•
P–10	1.8	50–600	C	**	C4–C5	LX	—	(K) •	—			5–40	Fs	•
P–11	1.1	50–600	C	**	C4–C5	LX	0	•	—			3–20	Fs, Ch	•
P–12	0.8	50–600	EA	**	C4	LX	•	+	—	*		5–250	Ch, Fs	0 •
P–13	1.5	60–500	EA	**	C4	LX	•	—	•	*	*	1–40	Fs, Sf	0 •

NOTES: % opaques is that of the total rock.
Shape: E = euhedral, EA = subhedral, A = anhedral, C = corroded, K = skeletal.
Very small opaques are numerous inclusions <1 micron (μ) in pyroxenes and feldspars; * means numerous; ** means very numerous.
Oxidation Index: C1 = unoxidized, C6 = highest oxidation state (see Wilson and Haggerty 1966).
Size = Size range of iron oxides in μ.
Mineral proportions: X = >50 %, + = 25–50%, 0 = 5–25 %, • = <5 %, — = absent; represent proportions of total opaques in this section.
Mag. il. inter. = Magnetite–ilmenite intergrowths. L indicates intergrowths are lamellar.
Presence of ulvöspinel is indicated by wormlike runnels in the magnetite portion of the magnetite–ilmenite intergrowths.
Presence of spinel is indicated by mosaic of transparent mineral, appearing black in polished section, within the magnetite portions of magnetite–ilmenite intergrowths.
Sulfides: Fs = pyrrhotite, Sf = pyrite, Ch = chalcopyrite.

truly representative of the other two. Nevertheless major ranking differences (e.g. site 10) would be difficult to account for by such explanations.

We suggest that this lack of correlation between the different criteria is because they are related in varying degree to three different types of stability. One is dispersive, a secondary component that adds a vector of random direction and size, such as a temporary component acquired if specimens are stored randomly in the earth's magnetic field. The second is directive, a secondary component contained by all specimens in the same direction, although not necessarily in the same amount. A viscous component in the earth's present field direction provides an example of this type. Because the coercivity of the directed component may vary from specimen to specimen it contains a dispersive element, but the dispersion will be along the great circle joining the primary and secondary components (e.g. Boesen *et al.*, Ginginbullen Dolerite, 1961). The third is very high intensity isothermal remanent magnetization caused by lightning strikes (Graham 1961): these are evident as high initial intensities giving scattered directions, with intensities falling off rapidly with magnetic cleaning. Sites 1, 7, and 10 of the Coronation Sills are examples.

Random components do not significantly affect the mean direction, but reduce the precision with which it can be defined. Directed components on the other hand, change the mean direction and must be removed before the true pole position can be calculated. It therefore seems to us that the end-point criterion, which is based on changes in direction, should be used to define the range of cleaning field (or temperature) to be used, and the optimum field within this range should be obtained by the precision criterion.

One polished section from each site (Table 5) was examined in an attempt to relate the magnetic properties to the opaque mineralogy. Although the percentage of opaque minerals ranges from 0.8 to 5.5, the similarities between sites are more marked than the differences. At all sites the predominant opaque mineral

is a coarse-grained lamellar intergrowth (presumably exsolution) of magnetite and ilmenite in an intermediate oxidation state (C3-C5, Wilson and Haggerty 1966). Anhedral, corroded, and skeletal grains are characteristic. All sections contain small sulfide grains in minor amounts; most common are pyrrhotite and chalcopyrite, followed by pyrite. All sections also contain individual grains of both magnetite and ilmenite, whereas hematite is either absent or very scarce. Very small opaque minerals included within the large pyroxene and feldspar grains may also be seen in all sections (see Hargraves and Young 1969). At some sites these were extremely common, but too small to be readily identified. Fine intergrowths of magnetite and ulvospinel and intergrowths of ilmenite and a transparent mineral, presumably spinel, were common in some sections.

We conclude that the wide range of within site precision is not caused mainly by petrographic differences but by differences in the proportions of coercivity spectra of isothermal remanent magnetization (IRM) and of viscous remanent magnetization (VRM), acquired at different sites because of different physical histories. The opaque mineral examination indicates that a limited range of mineralogical variation exists at all sites, and the magnetic tests show that the samples are in the intermediate stability range. Their magnetic properties are therefore delicate measures of small differences of the various types of stability within this range.

Paleomagnetic Directions

Results from 13 sites representing successive sills from south to north are given in Table 8, and detailed analysis of mean vectors from two sites (Holman Island site of Fig. 1 and site 14 of Fig. 2) are given in Table 7. The total time taken for cooling is unknown, but it is limited by two factors: first, no sill sampled is more than 140 m thick, hence the time required for the blocking temperature to pass from the edge to the center is small compared with that needed to complete a secular variation cycle (Jaeger and Green 1956); second, no reversals have been obtained from the Coronation Sills sequence, whereas Fahrig *et al.* (1971) obtained reversals in several units of their Franklin Magnetic Interval of which this is part. Hence the time taken to intrude the Coronation Sills must be less than that occupied by the Franklin Magnetic Interval as defined by Fahrig *et al.* (1971).

TABLE 6. Pilot alternating field demagnetization

Treatment[1]	N[2]	D[3]	I[4]	k[5]	α[5]	A.M.[6]
NRM	13	69	+25	5.2	20	1550
35	13	79	+23	5.8	19	966
70	13	79	+12	19	9.7	720
105	13	80	+ 3	23	8.9	481
210	13	83	− 5	14	11	305
280	13	88	− 7	10	14	175
400	13	83	−11	7.5	16	64.0
600	13	78	−19	2.7	32	29.7
1000	13	147	−23	1.3	70	24.3

[1]Treatment is the alternating magnetic field equivalent (see footnote 3 in text).
[2]N is the number of specimens, one from each site.
[3]D is the declination clockwise from true north.
[4]I is the inclination positive downwards.
[5]k is Fisher's (1953) best estimate of precision and α is the half-angle of the cone of confidence at P = 0.95.
[6]A.M. is the arithmetic mean intensity in e.m.u. $\times 10^{-6}$.

Three 32-mm diameter samples were taken from each of the main sites, P1-P13, and 8 samples from the detailed site, using a portable drill (Gaucher and Meilleur 1965). Five hand samples were taken at the Holman Island site. Two specimens were cut from each sample.

Alternating Field Demagnetization

One specimen from each of the 13 mainland sites (Fig. 1, P1-P13) was demagnetized in steps[4] using apparatus previously described (Irving *et al.* 1961; Roy, Reynolds, and Sanders 1971). The mean directions are shown in Fig. 4*e*. Minimum scatter for this pilot group (Table 6) occurs after cleaning in a peak field of 105 Oe, but an end-point is not quite reached, as shown by the movement of the mean direction away from the present field direction at higher fields.

No ideal end-point is evident although the direction changes between cleaning fields of 210 to 400 Oe are not significant (Fig. 4*e*). On the other hand both sites examined in detail (Site P-14 from the Couper Islands and the site at Holman Island, Table 7 and Fig. 4*d*) display good end-points between 200 and 400 Oe. One explanation is that the coercivity spectrum of

[4]Because the specimens were demagnetized along three mutually perpendicular axes and not tumbled, the value of the field used here was decreased by a factor of $2^{1/2}$, so that these results may be directly compared with those obtained by tumbling (Irving and Park, personal communication 1969).

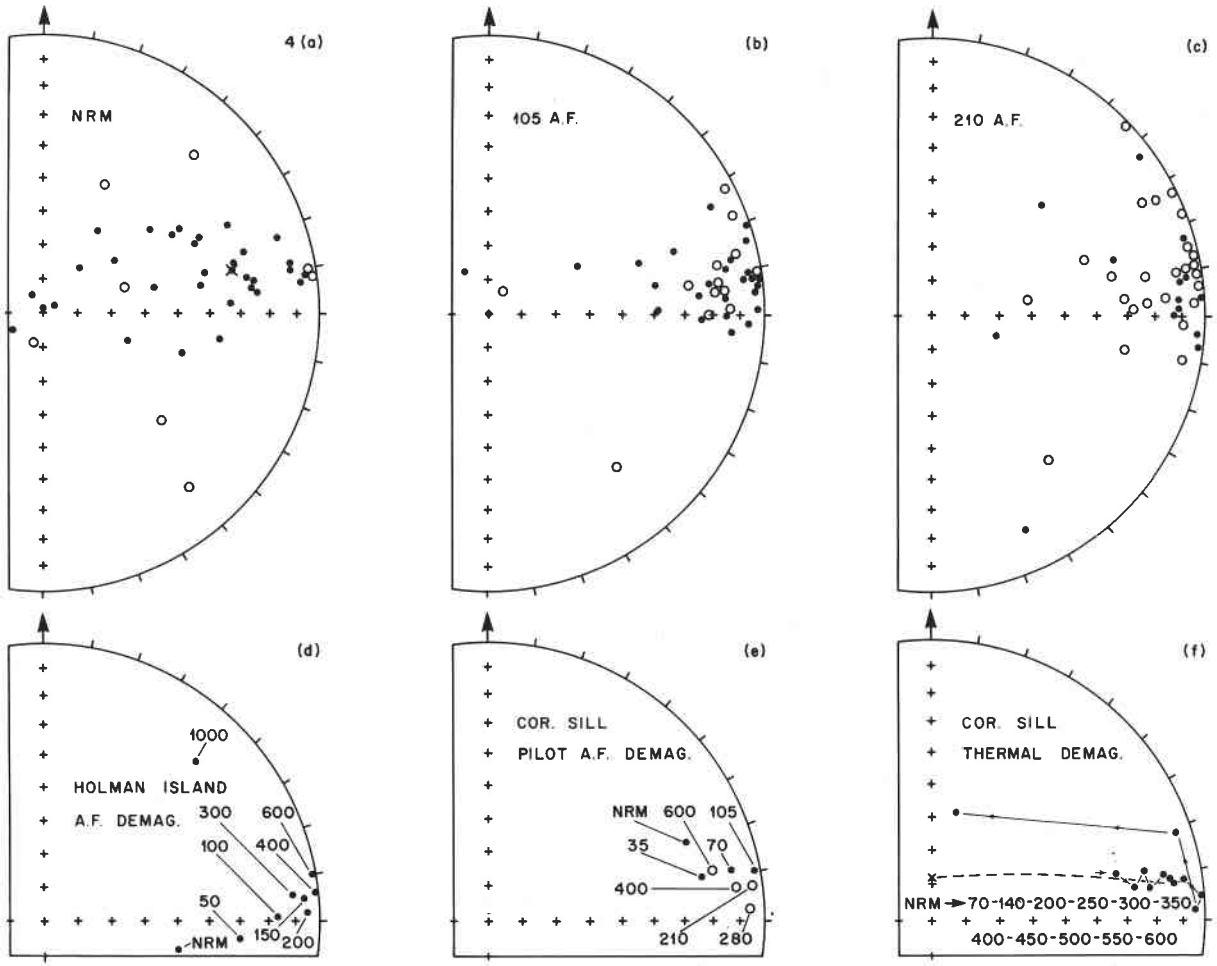


FIG. 4. (*a*), (*b*), and (*c*). Sample directions from mainland sites, initial and after 105 and 210 Oe peak alternating field (A.F.) treatments respectively. (*d*) A.F. demagnetization of the Holman Island site. (*e*) Mean direction of one specimen from each of mainland sites after subjection to peak A.F. shown. (*f*) Mean direction of one specimen from each mainland site after partial thermal demagnetization by heating to temperature shown and cooling in zero field. The present field direction is shown (x) and the great circle between this and the final A.F. cleaned direction is dashed. Open circles are on the upper hemisphere, and solid circles on the lower hemisphere of an equal-area net.

TABLE 7. Detailed site alternating field demagnetization*

Treatment	Site P–14[1]						Holman Island site[2]					
	N	D	I	k	α	A.M.	N	D	I	k	α	A.M.[3]
NRM	8	93	+12	4.1	3.1	941	5	102	+49	3.3	50	1040
50	—	—	—	—	—	—	5	95	+30	4.5	41	767
100	—	—	—	—	—	—	5	89	+17	8 7	29	547
150	—	—	—	—	—	—	5	85	+ 7	11.5	24	343
200	8	88	−15	16.4	14	304	4	88	+ 6	23.2	20	196
250	8	89	−16	14.3	15	216	—	—	—	—	—	—
300	8	91	−12	9.2	19	139	4	84	+11	27.8	18	81.3
400	8	90	−15	6.0	25	72.1	5	84	+ 2	45.5	12	44.9
600	8	144	−38	1.0	—	42.3	5	80	+ 2	15.0	20	10.9
800	8	277	− 5	2.0	53	52.8	—	—	—	—	—	—
1000	—	—	—	—	—	—	5	44	+22	2.0	76	5.4

*The treatment and symbols have the same significance as those in Table 6 except that the specimens (N) are all from the same site.

[1]We are grateful to A. Larochelle for the use of his biastatic magnetometer on which these measurements were made and to G. Freda, who made the measurements.

[2]We thank P.J. McKinnon for obtaining the 5 oriented samples from Holman Island during the course of other work.

[3]A.M. is the arithmetic mean intensity for the group of samples in e.m.u. $\times 10^{-6}$.

TABLE 8. Directions of magnetization after alternating field demagnetization*

Site	N.R.M.				105 Oe				210 Oe				M_{210}/M_o
	D	I	k	α	D	I	k	α	D	I	k	α	
P—1	105	− 7	0.90	—	92	−20	4.37	68	99	−22	4.63	65	.189
P—2	60	22	1.19	—	83	−28	2.72	97	93	−34	9.43	43	.178
P—3	77	39	10.9	39	82	8	51.7	17	87	2	100	12	.309
P—4	63	6	1.68	—	73	− 2	7.39	49	73	−16	19.1	29	.191
P—5	61	55	9.37	43	59	47	3.70	76	59	15	7.98	47	.021
P—6	72	65	5.70	57	—	—	—	—	75	39	3.36	82	.031
P—7	77	41	3.65	77	85	−10	26.9	24	87	−25	101	12	.235
P—8	71	31	76.9	14	75	6	106	12	62	0	29.3	23	.304
P—9	80	14	32.7	22	81	4	119	11	81	− 1	179	9	.383
P—10	81	− 7	1.63	—	87	− 2	55.5	17	82	2	61.0	16	.061
P—11	81	1	88.1	13	82	−18	88.4	13	80	−29	41.0	20	.302
P—12	95	48	23.3	26	87	31	30.9	23	107	15	4.28	69	.089
P—13	81	40	37.8	20	86	15	59.7	16	79	0	26.0	25	.098
Sites	78	27	9.67	14	81	6	9.97	14	82	− 4	11.3	13	.165
Samples	77	31	3.67	14	82	6	6.46	9.8	81	− 5	7.34	9.1	

*Symbols as for Table 6. M_{210}/M_o is the ratio of mean intensity, after the 210 cleaning, to initial intensity.

the viscous secondary component is different at different sites: where it is higher it has a coercivity approaching that of the primary component, which becomes much attenuated (with consequent scatter) during cleaning.

The site statistics in Table 8 show that at sites P-8, P-11, P-12, and P-13 the 105 Oe cleaning field yields the highest precision, whereas in general the within-site precision is higher after the 210 Oe cleaning. The sites with optimum precision after cleaning in 105 Oe do not relate to any measured opaque mineral characteristic, but are those in which the secondary components have a lower coercivity spectrum.

The pole position was calculated from the mean after 210 Oe cleaning (81°, + 4°), as this provided the highest precision in the end-point region.

Thermal Demagnetization

The ambiguity in the optimum magnetic field for cleaning the samples led to the need for careful thermal demagnetization. Therefore stepwise thermal demagnetization (e.g. Robertson 1969) of one specimen from each of the 13 mainland sites, was carried out in the non-magnetic oven built by Roy, Sanders, and Reynolds (1971), yielding the results of Table 9 and Figs. 4*f* and 6*b*).

While all the specimens contain a discrete

TABLE 9. Thermal demagnetization*

Discrete Group (7 Sites)									*Distributed Group (6 Sites)*								
Temp. (°C)	D	I	k	α	Lat	Long	dp	dm	Temp. (°C)	D	I	k	α	Lat	Long	dp	dm
NRM	77	+24	33.9	11	16.8 S	161.2	6	11	NRM	77	+46	1.95	67	30.4 S	155.3	55	86
70	79	+18	44.2	9	12.7 S	161.0	5	10	70	92	+46	1.79	73	24.7 S	140.9	61	94
140	79	+11	44.5	9	9.3 S	162.3	5	9	140	74	+46	1.92	68	31.6 S	157.9	56	88
200	80	+11	36.1	10	8.9 S	161.3	5	10	200	91	+41	2.44	55	21.1 S	144.2	40	67
250	80	+ 6	32.0	11	6.6 S	162.5	5	11	250	82	+33	2.63	52	19.6 S	154.9	33	59
300	80	+ 3	33.8	11	5.2 S	162.9	5	11	300	83	+31	2.66	51	18.9 S	154.0	33	58
350	81	+ 2	33.5	11	4.1 S	161.9	5	11	350	83	+31	2.91	48	17.8 S	154.4	30	53
400	83	− 2	30.2	11	2.0 S	161.4	6	11	400	81	+24	3.36	43	15.0 S	157.3	25	46
450	84	− 7	25.0	12	1.0 N	161.3	6	12	450	83	+19	4.02	38	11.5 S	157.0	21	40
500	79	− 5	22.9	13	1.6 S	165.5	6	13	500	83	+20	4.22	37	12.3 S	156.6	20	39
550	65	+ 5	6.1	27	11.7 S	177.1	13	27	550	80	+11	3.56	42	9.0 S	151.7	21	42
600	332	+31	1.5	27	35.9 S	278.3	48	87	600	65	+59	3.18	45	46.0 S	159.2	50	67

*Symbols as for Table 6. dp is the semi-axis of the oval of confidence at $P = 0.95$ passing through the site and the pole position, and dm is the semi-axis perpendicular to dp.

component with a blocking temperature above 500 °C, some specimens contain large components with blocking temperatures in the 20 to 500 °C range (Fig. 5*b*), and in other specimens these distributed components are very small (Fig. 5*a*). A notable feature of those specimens that have large distributed components is that their initial directions are strung towards the dipole field direction. Although the proportion of distributed component varies from specimen to specimen a distinction can be made between those in which it is large and those in which it is small. We have divided the specimens into two groups, those in which over half of the initial intensity is retained after the 500 °C heating, and those in which less than half the initial intensity remains. Mean directions and precisions have been calculated for each group (Table 9). Obvious differences between the two groups are tabulated (see text table).

Distributed Component Dominant	*Discrete Component Dominant*
1. Wide initial scatter.	1. Initially closely grouped.
2. Initial direction near dipole field direction.	2. Initial direction near Franklin direction.
3. Even steady movement away from dipole field direction from NRM–550 °C.	3. Small direction change between 140 and 500 °C.
4. Best direction, but highly scattered, at 550 °C. (80°, + 11°, $k = 3.6$).	4. Best direction 500 °C; scatter could be accounted for by secular variation. (79°, − 5°, $k = 22.9$).

Separation of the sites into the two groups with respectively large discrete and large distributed blocking temperatures shows that the group with a large discrete component has the higher coercivity spectrum (Fig. 5*c*).

It is not clear how the dipole field of today became imprinted on minerals with blocking temperatures in the 200–500 °C range. It is inconceivable that the rocks have recently been heated to this extent; therefore, unless the secondary component is an ancient one, perhaps acquired by heating caused by deep burial, in the same direction as the present dipole field, we must consider that those samples in which the distributed component is dominant were affected by some intrinsic physical change.

The mean direction of the discrete group after heating to 500 °C (79°, −5°) agrees very well with the mean from the bulk magnetic cleaning (81°, −5°) and is good supporting evidence for the belief that this was the direction of the magnetic field when the rocks cooled.

Discussion

Judged both from radiogenic ages (Table 1) and from the magnetization directions (Table 7) there is no reasonable doubt that the Coronation Sills formed during the Franklin Magnetic Interval of Fahrig *et al.* (1971). Both polarities were observed in this interval, although normally magnetized sites predominate, whereas all our Coronation Sills are reversely magnetized. Almost all of the initial site directions from Franklin rocks, including those reported here from the Coronation Sills, contain notable components along the present earth's dipole field direction (see Table 2 in Fahrig *et al.* 1971). If a small viscous component in a recent field direction was retained in the samples after cleaning, its effect on normally and reversely magnetized site poles would be opposite and it would be eliminated by averaging. Because there is no significant difference in pole position between the normally and reversely magnetized groups within the Franklin Magnetic Interval we conclude that viscous components, after cleaning, must be small.

We have added our 13 sites from the Coronation Sills to the 24 valid sites of Fahrig *et al* (1971) from the Franklin, yielding 19 normal and 18 reversed sites. The recalculated mean Franklin pole from these 37 sites is at 165.3° E, 4.8° N ($k = 26$, $\alpha_{95} = 4.7°$).

Conclusions

1. The Coronation Sills and their extrusive equivalents form a major plateau basaltic province of Hadrynian age. Assuming an average thickness of 1300 m they comprise about 165 000 km^3 of basaltic magna, comparable in volume with the 200 000 km^3 estimate of Kuno (1969) for the Columbia Plateau.

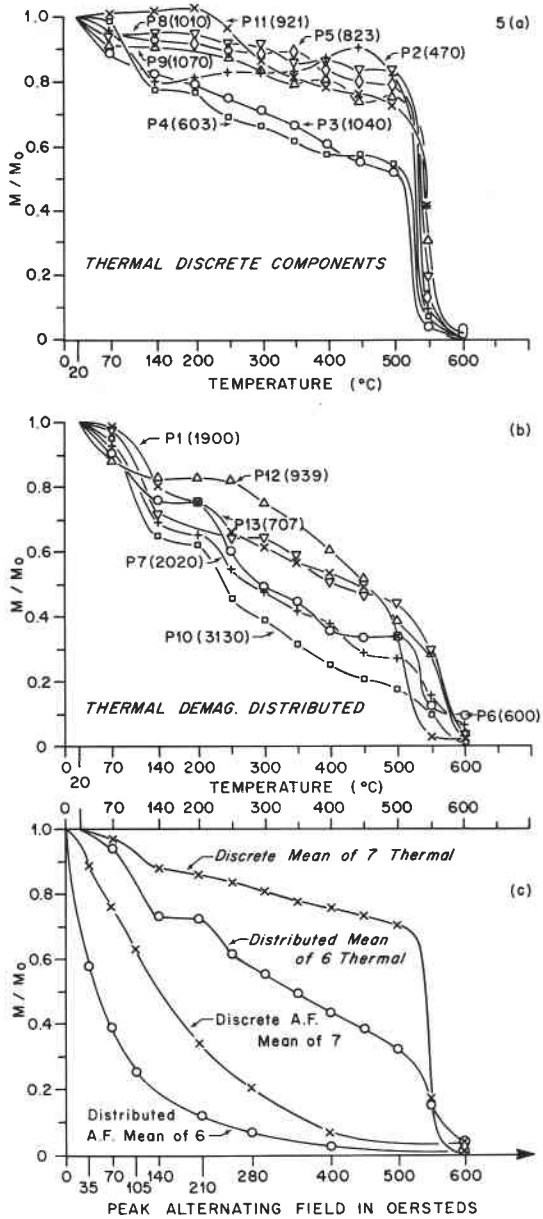


FIG. 5. (*a*) Thermal demagnetization curves for the specimens with large discrete components. The numbers in brackets are the n.r.m. intensities in e.m.u. $\times 10^{-6}$. (*b*) Thermal demagnetization curves for the specimens with large distributed components. (*c*) Mean thermal and alternating field demagnetization curves of the two groups with respectively predominantly discrete and predominantly distributed components.

2. The sills were presumably fed by dikes of Franklin age (Fahrig *et al* 1971) and intruded during the Franklin Magnetic Interval.

3. The magma from which the sills consolidated was a saturated to slightly undersaturated, almost completely undifferentiated, tholeiitic, basaltic magma.

4. The Coronation Sills are less oxidized and contain less potash and titania and more lime than the nearby but older Coppermine River lavas (Fig. 3).

5. The sills are higher in potash than tholeiitic basalts of oceanic or geosynclinal environments; they are chemically very similar to the Karroo dolerite (Table 3).

6. Initial directions of magnetization are quite scattered, but the scatter is reduced by magnetic cleaning in a peak field of 210 Oe (Fig. 4) and less effectively by thermal cleaning to 500 °C. The scatter is attributed to lightning strikes, temporary components acquired in the laboratory, viscous components acquired in the earth's magnetic field, secular variation, and measuring error.

7. All sites are reversely magnetized (directions low negative east) and are thought to reflect a reversed ambient field at the time of intrusion.

8. A very high proportion of the opaque minerals are large lamellar intergrowths of magnetite and ilmenite, which are considered to dominate the magnetic properies of the rock.

9. Using the precision of a group of directions from samples as the criterion, the optimum magnetic cleaning field is 105 Oe for some sites and 210 Oe for others (Table 8). At most sites the mean direction moves farther from the present field direction between the 105 Oe and 210 Oe treatments, and slightly farther again in fields up to 400 Oe, above which the scatter is very high (e.g. Table 7). A field of 210 Oe, a compromise between minimum scatter and minimum movement between two cleanings, gives good between-site grouping.

10. Thermal demagnetization separates the sites into two groups. One group has a large discrete blocking temperature above 500 °C and is characterized by small dispersion. The other group has blocking temperatures distributed between 20 and 550 °C, and displays a distribution strung out along a great circle.

11. The pole position calculated from the mean of 13 sites after magnetic cleaning is at 1° S, 163° E, (dp = 6, dm = 13).

12. Combining the directions from these 13

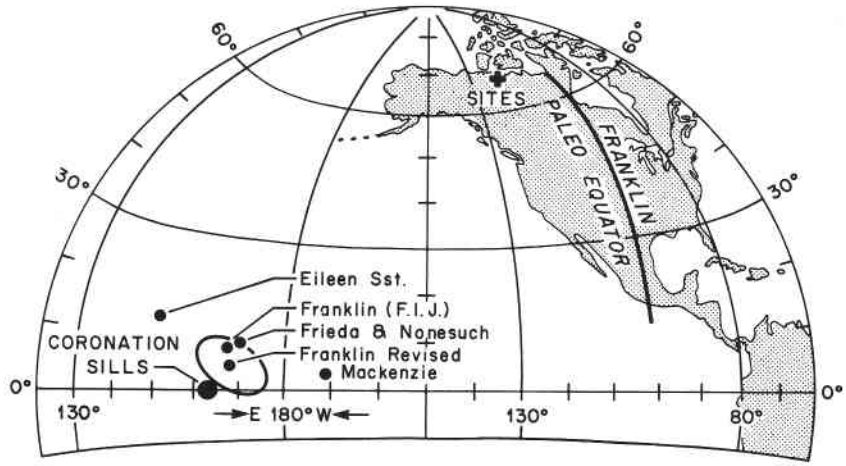


FIG. 6. An equal-area projection showing the position of the mean Coronation Sill pole relative to North America. Also shown are the Eileen Sandstone and Frieda and Nonesuch poles of DuBois (1962), the mean Mackenzie Magnetic Interval pole of Fahrig and Jones (1969), the Franklin Magnetic Interval pole of Fahrig *et al.* (1971), and the revised Franklin Magnetic Interval pole incorporating our sites from the Coronation Sills, with the $P = 0.95$ oval of confidence.

TABLE 10. Magnetic and related properties

Site	Sill No.[1]	n.r.m.[2] $\times 10^{-3}$	m.d.f.[5] n.r.m.	a.r.m.[3] $\times 10^{-3}$	m.d.f.[5] a.r.m.	i.r.m.[4] $\times 10^{-3}$	m.d.f. i.r.m.	Fe_2O_3/ FeO	Total Fe as FeO*	Opaques	lam.[6] int.	Blocking Temp. (°C)[7]
P—1	1	3.53	60	2.25	225	890	305	0.343	10.99	5.4	0.94	D & 520
P—2	2	.96	75	—	—	520	290	0.382	11.51	2.4	0.88	520
P—3	2	.77	240	—	—	1075	335	0.316	12.79	1.6	0.81	520 & D
P—4	3	.45	295	—	—	840	320	0.288	12.88	4.1	0.93	520 & D
P—5	5	1.34	185	—	—	1265	280	0.311	14.64	5.3	0.94	520
P—6	6	1.01	130	—	—	770	270	—	—	5.5	0.95	D & 520
P—7	8	2.97	45	2.21	335	820	345	0.244	13.48	4.4	0.80	D &(520)
P—8	8	1.09	260	2.60	240	1113	340	0.279	12.80	3.3	0.91	520
P—9	9	1.86	135	3.03	275	1490	445	0.219	11.53	2.9	0.90	530
P—10	10	3.47	30	—	—	380	175	0.336	11.02	1.8	0.89	D &(560)
P—11	11	1.09	160	2.03	205	775	290	0.296	12.23	1.1	0.91	540 &(D)
P—12	13	1.41	90	—	—	1490	245	0.635	21.46	0.8	0.63	530 &(D)
P—13	14	.28	140	.89	200	323	295	—	—	1.5	0.93	D & 550

*Classical analyses performed in the Geol. Surv. Can. analytical laboratories by L. Seymour; one sample from each site.

[1]The sills are numbered from the bottom so that No. 1 is the farthest south and probably the oldest.

[2]The n.r.m. is the natural remanent magnetization in electromagnetic units.

[3]The a.r.m. is the anhysteritic magnetization acquired in an alternating field of 2500 Oe and a bias steady field of 0.3 Oe.

[4]The i.r.m. is the isothermal remanent magnetization induced in 11 000 Oe.

[5]The m.d.f. of n.r.m. and a.r.m. is the field in oersteds required to reduce the initial intensity by half. In the case of i.r.m. the m.d.f. is the field in oersteds required to reduce the mean intensity to zero: in this case the lower coercivity grains are magnetized oppositely to the higher coercivity grains. The number in the opaques column is the percentage of opaque minerals in the polished section.

[6]The number in the column headed "lam. int." is the fraction of the opaque minerals in the polished section that consists of lamellar intergrowths.

[7]Under blocking temp. D represents widely distributed components. The figures represent the temperature in °C. The predominant blocking temperatures are first and minor ones are bracketed.

Coronation Sills sites with the 24 sites from the Franklin Magnetic Interval obtained by Fahrig *et al.* (1971), the revised Franklin pole position is 4.8° N, 165.3° E. ($\alpha_{95} = 4.7°$).

Acknowledgments

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